

Solutions for Algebraic Geometry Hartshorne

Guo Haoyang

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1 Varieties

2 Schemes

2.1 Sheaves

2.2 Schemes

Question 2.1

Let A be a ring and $X = \text{Spec} A$, let $f \in A$ and $D(f) := X \setminus V(\langle f \rangle)$. Show that

$$(D(f), \mathcal{O}_X|_{D(f)}) \cong (\text{Spec} A_f, \mathcal{O}_{\text{Spec} A_f})$$

Question 2.2

Let $(X, \mathcal{O}_X)$ be a scheme and $U \subseteq X$ be an open subset. Show that the *induced scheme* on U , $(U, \mathcal{O}_X|_U)$ is a scheme.

Question 2.3

- (1) Show that $(X, \mathcal{O}_X)$ is reduced $\Leftrightarrow \forall P \in X$, the local ring $\mathcal{O}_{X,P}$ has no nilpotent element.
- (2) Let $(X, \mathcal{O}_X)$ be a scheme and $(\mathcal{O}_X)_{\text{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\text{red}}$, where for any ring A , $A_{\text{red}} := A/I$ with I the ideal of nilpotent elements of A . Show that the *reduced scheme* $(X, (\mathcal{O}_X)_{\text{red}})$ is a scheme and there is a morphism of schemes $(X, (\mathcal{O}_X)_{\text{red}}) \rightarrow (X, \mathcal{O}_X)$, which is a homeomorphism on the underlying topological space.
- (3)

Question 2.4

Given a morphism $f : X \rightarrow \text{Spec} A$, and a associated map on sheaves $f^\# : \mathcal{O}_{\text{Spec} A} \rightarrow f_* \mathcal{O}_X$, we obtain a homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$ by taking global sections.

Thus, there is a natural map

$$\alpha : \text{Mor}_{\mathbf{Sch}}((X, \mathcal{O}_X), (\text{Spec } A, \mathcal{O}_{\text{Spec } A})) \rightarrow \text{Mor}_{\mathbf{CRing}}(A, \Gamma(X, \mathcal{O}_X)) \quad a$$

Show that α is a bijection map.

^aHom is synonymous to Mor.

Proof:

□

Question 2.5: Final objects in category of schemes

Describe $\text{Spec } \mathbb{Z}$. Show that $\text{Spec } \mathbb{Z}$ is a final object for the category of schemes.

Proof: Let $(X, \mathcal{O}_X)$ be an arbitrary scheme. By definition of schemes, $\exists$ an open cover of X , $\{U_i\}_{i \in I}$, such that $\forall i$, $(U_i, \mathcal{O}_X|_{U_i})$ is an affine scheme. i.e. $\forall i$, $\exists R_i$ a ring, such that $(U_i, \mathcal{O}_X|_{U_i}) \cong (\text{Spec } R_i, \mathcal{O}_{\text{Spec } R_i})$.

Since $\mathbb{Z}$ is initial in the category $\mathbf{CRing}$, then $\forall i$, $\exists!$ ring homomorphism $\mathbb{Z} \rightarrow R_i$ which induces unique morphism of affine schemes $(\text{Spec } R_i, \mathcal{O}_{\text{Spec } R_i}) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$. Hence, there is a unique morphism $(U_i, \mathcal{O}_X|_{U_i}) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$ by composing the above-mentioned isomorphism.

This result lifts to the unique morphism of schemes $(X, \mathcal{O}_X) \rightarrow (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}})$ because the Hom of scheme, $\mathcal{H} := \text{Hom}_{\mathbf{Sch}}((X, \mathcal{O}_X), (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}}))$ is a sheaf.

$$\mathcal{H}(U) := \text{Hom}_{\mathbf{Sch}}((U_i, \mathcal{O}_X|_{U_i}), (\text{Spec } \mathbb{Z}, \mathcal{O}_{\text{Spec } \mathbb{Z}}))$$

(So, the locality and gluing axioms extend morphisms on $(U_i, \mathcal{O}_X|_{U_i})$ globally.)

□

Question 2.6: Initial objects in category of schemes

Describe $\text{Spec } \{0\}$. Show that $\text{Spec } \{0\}$ is an initial object for the category of schemes.

Question 2.7: Characterisation of tangent spaces

Let $(X, \mathcal{O}_X)$ be a scheme over a field k and $k[\epsilon]/\langle \epsilon^2 \rangle$ be the ring of dual numbers over k . Show that giving a k -morphism $\varphi : \text{Spec } k[\epsilon]/\langle \epsilon^2 \rangle \rightarrow X \Leftrightarrow$ giving $x \in X$, rational over k and an element of Zariski tangent space T_x .

Remark Some analysis on $k[\epsilon]/\langle \epsilon^2 \rangle$:

(1) When $k = \mathbb{R}$, $\forall f(x) = \sum_i a_i x^i \in (\mathbb{R}[\epsilon]/\langle \epsilon^2 \rangle)[x]$, $f(\overline{a + b\epsilon}) = f(\overline{a}) + \overline{b}f'(\overline{a})\overline{\epsilon}$.

(2) For general fields k , units are elements of the form $a + \langle \epsilon^2 \rangle$ or $a + b\epsilon + \langle \epsilon^2 \rangle$ with $a, b \neq 0$. So, non-units of $k[\epsilon]/\langle \epsilon^2 \rangle$ are of the form $b\epsilon + \langle \epsilon^2 \rangle$ with $b \neq 0$, which are in $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$.

Since every proper ideal cannot have any units, so it is a subset of the set of non-units. Hence a subset of $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$. Therefore, $\langle \epsilon \rangle/\langle \epsilon^2 \rangle$ is the unique maximal ideal of $k[\epsilon]/\langle \epsilon^2 \rangle$. Thus, $k[\epsilon]/\langle \epsilon^2 \rangle$ is local.

Proof: $\Rightarrow$ Consider we have a morphism $\varphi : \text{Spec } k[\epsilon]/\langle \epsilon^2 \rangle \rightarrow X$. Then, x can be given by the point that the unique maximal ideal of $k[\epsilon]/\langle \epsilon^2 \rangle$ goes, i.e. $x := \varphi(\langle \epsilon \rangle/\langle \epsilon^2 \rangle)$ $\square$

Question 2.8: Anti-equivalence of Fld and Sch

The category of field **Fld** is antiequivalent to the category of scheme **Sch**.

Question 2.9: Generic points

Question 2.10: $\text{Spec } \mathbb{R}[x]$

Question 2.11: $\text{Spec } \mathbb{F}_p[x]$

Question 2.12: Glueing lemma

Question 2.13: Quasi-compactness of a topological space

- (1) Show that a topological space is Noetherian $\Leftrightarrow$ every *open subset* is quasi-compact.
- (2) Let $(X, \mathcal{O}_X)$ be an affine scheme. Show that X is quasi-compact but not necessarily Noetherian.
- (3) If A is a Noetherian ring, then $\text{Spec}(A)$ is a Noetherian topological space.
- (4) The converse of (3) is not always true. Give an example (A is not Noetherian ring but $\text{Spec}(A)$ is a Noetherian topological space).

Question 2.14: Projective scheme

- (1) Let S be a graded ring. Show that $\text{Proj } S = \emptyset$ iff $\forall x \in S_+, x$ is nilpotent.
- (2) Let $\varphi : S \rightarrow T$ be a graded homomorphism of graded rings. $U := \{[\mathfrak{p}] \in \text{Proj } T \mid \varphi(S_+) \not\subseteq \mathfrak{p}\}$. Show that U is an open subset of $\text{Proj } T$ and φ determines a natural morphism $f : U \rightarrow \text{Proj } S$.
- (3) The morphism f can be an isomorphism even when φ is not an isomorphism. Show the following example:
Let d_0 be an integer. $\forall d \geq d_0, \varphi_d : S_d \rightarrow T_d$ is an isomorphism, then $U = \text{Proj } T$ and the morphism $f : \text{Proj } T \rightarrow \text{Proj } S$ is isomorphic.
- (4) Let V be a projective variety with homogeneous coordinate ring S . Show that $t(V) \cong \text{Proj } S$.

Proof: (1) $\Rightarrow$ Suppose $\exists y \in S_+$ that is not a nilpotent. Then, we will show that there is a homogeneous prime ideal that satisfies $S_+ \not\subseteq$ this ideal.

$y = \sum_{i_1 \leq d \leq i_m} y_d$ with each $y_d \in S_d$. Then, y is not a nilpotent implies that at least one y_t in the sum is not a nilpotent. Let $T := \{1, y_t, y_t^2, \dots\}$. Consider the localization of $T^{-1}S =: S_{y_t}$. Since y_t is not a nilpotent, $0 \notin T$ and S_{y_t} is not a zero ring, which implies that the degree-0 localization $S_{(y_t)}$ is not a zero ring.¹ Furthermore, $\text{Spec } S_{(y_t)}$ is non-empty because every non-zero ring has at least one maximal ideal. The following isomorphism

$$D_+(y_t) \cong \text{Spec } S_{(y_t)}$$

implies that $D_+(y_t) \neq \emptyset \Rightarrow \text{Proj } S \neq \emptyset$. ($D_+(y_t) \subseteq \text{Proj } S$)

$\Leftarrow$ Suppose that $\forall x \in S_+, x$ is nilpotent. We will show that there is no homogeneous prime ideal $\mathfrak{p}$ that $S_+ \not\subseteq \mathfrak{p}$. Let $\mathfrak{p}$ be a homogeneous prime ideal. $\forall x \in S_+$, since $\exists n, x^n = 0 \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$ because $\mathfrak{p}$ is prime, implying $S_+ \subseteq \mathfrak{p}$ for all homogeneous prime ideal $\mathfrak{p}$.

(2)

□

Question 2.15

- (1) Let V be a variety over the algebraically closed field k . Show that $P \in t(V)$ is a closed point $\Leftrightarrow$ the residue field of P is k .

Question 2.16

Let $(X, \mathcal{O}_X)$ be a scheme and $f \in \Gamma(X, \mathcal{O}_X)$ be a global section. $\mathfrak{m}_x$ is the maximal ideal of the local ring $\mathcal{O}_{X,x}$. $X_f := \{x \in X : f_x \notin \mathfrak{m}_x\}$ where $f_x \in \mathcal{O}_{X,x}$ or a germ of the function f .

- (1) Suppose $(U, \mathcal{O}_X|_U)$ is an open affine scheme of $(X, \mathcal{O}_X)$ with $U = \text{Spec } B$. $f|_U \in \Gamma(U, \mathcal{O}_X|_U) \cong B$. Show that $X_f \cap U = D(f|_U)$. Conclude that X_f is an open subset of X .
- (2) Let $(X, \mathcal{O}_X)$ be quasi-affine and $a \in \Gamma(X, \mathcal{O}_X)$ such that $a|_{X_f} = 0$. Show that $\exists n > 0, f^n a = 0$.

Proof: (1) $U \cap X_f = \{x \in U \mid f_x \notin \mathfrak{m}_x\}$

(2) Let

□

Question 2.17: A criterion for affineness

Question 2.18: Properties of a ring homomorphism to the induced morphisms

- (1) Let A be a ring, $X = \text{Spec } A$ and $f \in A$. f is nilpotent $\Leftrightarrow D(f) = \emptyset$.
- (2) Let $\varphi : A \rightarrow B$ be a homomorphism of rings and let $f : Y = \text{Spec } B \rightarrow \text{Spec } A = X$ be the induced morphism of affine schemes. φ is injective $\Leftrightarrow f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$.
- (3) φ is surjective $\Leftrightarrow f : Y \rightarrow C$ ^a is a homeomorphism and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective. Hence,

^a C is a closed subset of X

Proof: (1) Suppose that f is a nilpotent, then $\exists n$, such that $f^n = 0$. This means that $\forall [\mathfrak{p}] \in \text{Spec } A, f^n \in \mathfrak{p} \Rightarrow f \in \mathfrak{p}$.

$$D(f) = \{[\mathfrak{p}] \in \text{Spec } A : f^n \notin \mathfrak{p}\} = \emptyset$$

¹This is true for all S_y such that $y \in S_d$ with $d \geq 1$.

$D(f) = \emptyset \Leftrightarrow \forall$ prime ideal $\mathfrak{p}$, $f \in \mathfrak{p}$. $\{0\}$ is also a prime ideal. Thereby, $f = 0$ and f is a nilpotent.

□

Question 2.19

Let A be a ring commutative?. Show TFAE:

- (1) $\text{Spec } A$ is disconnected.
- (2) $\exists$ non-zero orthogonal idempotents $e_1, e_2 \in A$, i.e. $e_1^2 = e_1$, $e_2^2 = e_2$, $e_1 e_2 = 0$ and $e_1 + e_2 = 1$.
- (3) $A \cong A_1 \times A_2$, where A_1, A_2 are two non-zero rings.

2.3 First properties of schemes

3 Cohomology

4 Curves

5 Surfaces